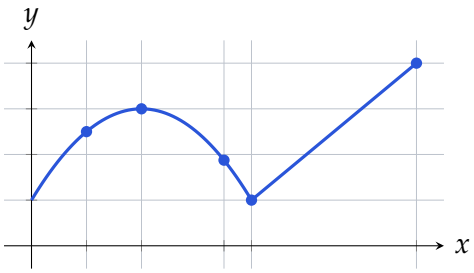
Lesson 1-8 • MVT

Notes (filled in)

Objective: recognize when the Mean Value Theorem applies, and use it.

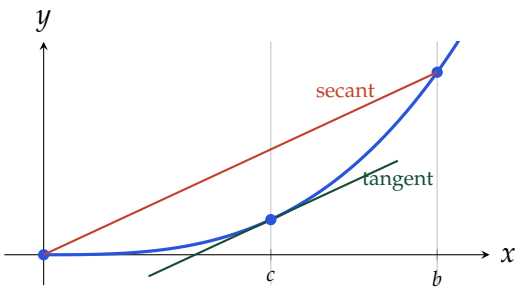
Review: differentiability

What is the instantaneous rate of change at each marked point: positive, negative, 0 or DNE?



$x = 1$: positive
$x = 2$: 0
$x = 3.5$: negative
$x = 4$: DNE (corner)
$x = 7$: DNE (endpoint)

The Mean Value Theorem (MVT)



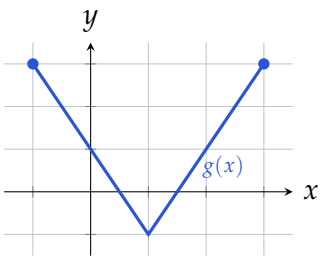
Needs	If f is continuous on $[a,b]$ and differentiable on (a,b),
Proves	then there is a c in (a,b) with $f'(c) = \dfrac{f(b)-f(a)}{b-a}$.

In words: the slope of the tangent at some point equals the slope of the secant (the average rate of change). The MVT is about slopes.

Example 1

f is a differentiable function with $f(2) = 5$ and $f(7) = 20$. Explain why there must be a value c, $2 < c < 7$, with $f'(c) = 3$.

f is differentiable, so it is also continuous on $[2,7]$: the MVT applies. $\dfrac{f(7)-f(2)}{7-2} = \dfrac{20-5}{5} = 3$, so there is a c in $(2,7)$ with $f'(c) = 3$.

Example 2	 $g(x) = 2\|x-1\| - 1$ on $[-1,3]$. Find the average rate of change of g on $[-1,3]$. Does the MVT guarantee a c in $(-1,3)$ with $g'(c)$ equal to it? $\frac{g(3)-g(-1)}{3-(-1)} = \frac{3-3}{4} = 0$. No: g is not differentiable at $x = 1$ (sharp corner), so the MVT does not apply. In fact no point has slope 0: the slopes are only -2 and 2.
Summary	MVT: if f is continuous on $[a,b]$ and differentiable on (a,b), some tangent slope equals the average rate of change $\frac{f(b)-f(a)}{b-a}$. Always check differentiability first: one corner is enough to break it.